METRIC

MIL-DTL-32613(SH)

11 September 2018

DETAIL SPECIFICATION

CONTROLLER, AUXILIARY-SYSTEM, NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers alternating current (AC) and direct current (DC) shipboard controllers that can be used for a variety of auxiliary systems, to include non-programmable controllers and programmable logic controllers (PLC) requiring cybersecurity measures.

1.2 Classification. Controllers in this specification are of the following types:

a. Type NO – Non-programmable (mechanical, magnetic, and others)

b. Type PP – PLC-based programmable

1.3 Part or identifying number (PIN). PINs for these auxiliary controllers are created as follows:

M	**32613**	**-**	**XX**	**X**	**X**	**X**
Prefix for military specification	Specification number	-	Type (see code below)	Duty (see code below)	Power (see code below)	Size (see code below)

Example: M32613-PP1AF is a PLC-based, continuous duty, 115-volt AC (VAC), size 2 controller.

Type		**Duty**		**Power**		**Size**	
PIN code	**Controller**	**PIN code**	**Duty**	**PIN code**	**Size**	**PIN code**	**Size**
NO	Non-programmable	1	Continuous	A	115 VAC, three-phase	Per table II	Per table II
PP	PLC-based	2	Special	B	440 VAC, three-phase		
				C	24 VDC		
				D	115 VDC		
				E	230 VDC		
				F	250 VDC		

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AMSC N/A FSC 6110

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59125 - Terminal Boards, Molded, Barrier Screw and Stud Types and Associated Accessories

A-A-59486 - Padlock Set (Individually Keyed or Keyed Alike)

A-A-59487 - Padlock (Key Operated)

A-A-59781 - Light Emitting Diodes for use as Indicator Lights

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-DTL-3661 - Lampholders, Indicator Lights, Indicator Light Housings, and Indicator Light Lenses, General Specification for

MIL-L-3661/63 - Lampholder, Lights, Indicator (Housing), Style LH96

MIL-L-3661/65 - Lampholder, Lights, Indicator (Housing), Style LH98

MIL-PRF-8805 - Switches and Switch Assemblies, Sensitive, Snap Action (Basic, Limit, Push Button and Toggle Switches), General Specification for

MIL-DTL-15109 - Resistors and Rheostats, Naval Shipboard

MIL-DTL-15291 - Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for

MIL-DTL-18396 - Switches, Meter and Control, Naval Shipboard

MIL-R-19523 - Relays, Control

MIL-DTL-21604 - Switches, Rotary, Multipole and Selector, General Specification for

MIL-PRF-24711 - Switch, Proximity, Solid-State

MIL-PRF-32168 - Variable Speed Drive System for Induction and Synchronous Machines

MIL-PRF-32407 - Mounts, Resilient (Surface Ship Application)

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-464	-	Electromagnetic Environmental Effects Requirements for Systems
MIL-STD-740-2	-	Structureborne Vibratory Acceleration Measurements and Acceptance Criteria of Shipboard Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1310	-	Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1399-300	-	Section 300B, Electric Power, Alternating Current
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits
MIL-STD-2042	-	Fiber Optic Cable Topology Installation Standard Methods for Surface Ships and Submarines

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-454	-	General Guidelines for Electronic Equipment
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume I

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMITTEE ON NATIONAL SECURITY SYSTEMS INSTRUCTION (CNSSI)

CNSSI 1253	-	Security Categorization and Control Selection for National Security Systems

(Copies of this document are available online at https://www.cnss.gov/CNSS/issuances/Instructions.cfm.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9073-A2-HBK-010	-	Installation and Inspection Information; Resilient Mount Handbook
S9086-KC-STM-010/300	-	Electric Plant-General
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
T9070-BR-DPC-010/202-1	-	Design Practices and Criteria for the Architecture of Shipboard Machinery Control Systems

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(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM F1166 - Standard Practice for Human Engineering Design for Marine Systems, Equipment, and Facilities

(Copies of this document are available online at www.astm.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61131 - Programmable Controllers (All Parts)

(Copies of this document are available online at http://webstore.iec.ch.)

IPC

J-STD-001 - Requirements for Soldered Electrical and Electronic Assemblies

(Copies of this document are available online at www.ipc.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Control and Systems: General Requirements

NEMA 250 - Enclosures for Electrical Equipment (1000 Volts Maximum)

(Copies of these documents are available online at www.nema.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Controllers furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.4).

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Construction and materials. Unless otherwise specified herein, controllers shall conform to the requirements of MIL-DTL-917 for the following items:

a. Materials
b. Electrical shock hazard
c. Electrical creepage and clearance distances
d. Threaded parts and fastening devices
e. Electrical insulation
f. Insulating procedures

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g. Wire, wiring methods, and marking

h. Parts

i. Processes

3.3.1 Prohibited materials. Controllers shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2.

3.4 Environmental conditions. Unless otherwise specified (see 6.2), auxiliary controllers shall be operable in temperatures ranging from -40 to 60 degrees Celsius (°C) (-40 to 140 degrees Fahrenheit [°F]) with relative humidity from 0 to 100 percent, noncondensing.

3.4.1 Noise. Controllers shall meet the structure-borne noise requirements for type III equipment of MIL-STD-740-2, and airborne noise requirements for grades A3 and A12 equipment of MIL-STD-1474. Snubbers and mounts shall be in accordance with S9073-A2-HBK-010 or MIL-PRF-32407.

3.4.2 Shock. Equipment shall be able to withstand grade A, class 1, type A shock acceptance requirements of MIL-DTL-901 without change of operational state, mechanical or electrical damage, or electrical malfunction. Equipment shall not fail to function in any operational mode as a result of the shock test. Resilient mounts inside or on the controller enclosure may be utilized to meet this requirement. If used, resilient mounts shall be in accordance with S9073-A2-HBK-010 or MIL-PRF-32407. Contact performance during impact shall be as follows:

a. AC line contacts. When shock tested as specified in 4.6.11, closed line contacts of contactors shall not open for more than 200 milliseconds. Open line contacts shall not close for more than 20 milliseconds.

b. DC line contacts. When shock tested as specified in 4.6.11, closed line contacts of contactors shall not open for more than 500 milliseconds. Open line contacts shall not close for more than 75 milliseconds.

c. Associated switching device contacts. Closed contacts may open, or open contacts may close, for a duration of not more than 10 milliseconds for all pressure and temperature switches as well as for timing relays that are designed for vital applications. Unless otherwise specified (see 6.2), closed contacts may open, or open contacts may close, for a duration of not more than 20 milliseconds for the other relays and switches designed for non-vital applications. The equipment shall not alter state when tested as specified in 4.6.11.

d. Auxiliary contacts. Closed contacts shall not open more than 20 milliseconds. Unless otherwise specified (see 6.2), open contacts shall not close for more than 20 milliseconds. Test conditions shall be as specified in 4.6.11.

3.4.3 Vibration. Equipment shall be able to withstand type I and type II vibration requirements of MIL-STD-167-1 without mechanical damage, contact chatter, transfer, time delay relay, temperature or pressure switch set point change, or other electrical malfunctioning. Resilient mounts inside the controller enclosure may be utilized to meet this requirement. If used, resilient mounts shall be in accordance with S9073-A2-HBK-010 or MIL-PRF-32407.

3.4.4 Electromagnetic interference (EMI) and electrical bonding and grounding. Controllers shall comply with the EMI limits and criteria of MIL-STD-461, and shall be bonded and grounded in accordance with MIL-STD-461, MIL-STD-464, and MIL-STD-1310 to minimize adverse EMI effects and to protect personnel from electrical shock (see 3.10).

3.4.5 Inclination. Controllers shall operate when permanently inclined in any direction, at any angle up to 30 degrees from the normal operating position. Controllers shall be constructed to function without damage during temporary inclinations, such as those produced by a ship’s rolling of 60 degrees in any direction from the normal operating position.

3.4.6 Electrical service conditions. When specified (see 6.2), controllers shall operate properly under the electrical service conditions specified in MIL-DTL-2212 and herein, to include steady-state voltage and frequency; transient voltage and frequency; spike voltage; power interruption; leakage current, output leakage current, and output voltage drop; and voltage ramp rate.

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3.5 Design. Auxiliary controllers shall consist of electrical and electronic components, contactors, and related circuitry contained in an enclosure. Controllers shall be non-programmable or programmable (see 6.2). Controllers shall be capable of being operated by a supervisory machinery control system or a human operator, locally or remotely.

3.5.1 Types of auxiliary controllers.

3.5.1.1 Non-programmable controllers. Electromechanical, magnetically operated non-programmable controllers shall be used for direct control of equipment. These shall have no programmable or remotely changeable operation characteristics.

3.5.1.2 Programmable controllers. Programmable controllers shall have the capability to have operational characteristics and monitoring parameters programmed using a PLC. PLCs shall be in accordance with IEC 61131 and shall be capable of performing normal programmed functions without human operator action. All programmable controllers shall:

a. Be capable of implementing the cybersecurity measures of 3.5.4.

b. Be capable of operating in a network or as a standalone controller.

3.5.2 Power.

3.5.2.1 AC controllers. AC controllers shall operate on type I, 60-hertz (Hz) power as specified in MIL-STD-1399-300 and shall operate between 80 and 110 percent of rated voltage at the specified ambient temperature (see 3.4). All AC controllers shall be three-phase.

3.5.2.2 DC controllers. DC controllers of 115 and 230 volts DC (VDC) (nominal) shall operate between 80 and 110 percent of rated voltage at the specified ambient temperature. DC controllers of 250 VDC (nominal) shall operate between 180 and 355 VDC at the specified ambient temperature (see 3.4). DC controllers of 24 VDC shall operate between 12 and 32 VDC.

3.5.3 Circuit arrangement. Circuit arrangements shall be as follows:

a. Controllers shall operate on an ungrounded system.

b. Each controller shall be electrically independent, except for interlocks, and will normally derive its power from the same source as the controlled device. Control circuits shall be protected by fuses upstream of all control components, including transformers.

c. A fault in one controller shall not disable any non-interlocked controller.

d. Current transformer secondaries shall be grounded to the controller enclosure.

e. Main line fuses, disconnects, or circuit breakers shall not be provided as part of the controller.

f. Contacts shall not be connected in parallel to obtain higher current carrying capacity. Line contacts of one contactor shall not be used to maintain power to another contactor unless tests (see 4.6.12) have been completed to ensure that a shock will not cause the maintained contactor to drop out.

g. Multi-speed and reversing controllers shall be arranged so that speed and direction changes are affected by a single operation. Controllers shall be constructed with safeguards to ensure it is not possible for a motor to be in a different mode of operation than indicated by the associated switching device's identification plate. Multi-speed and reversing controllers shall use electrical and mechanical interlocks to prevent attempted operation in more than one mode.

h. When timing relays are used to bypass safety switches during the starting period, the circuit shall be wired so that an opening in the relay circuit will prevent starting.

i. The stop device of a controller shall be interlocked such that the controlled device will be de-energized and can be started again only upon actuating the start device. A controller designed for low voltage release (LVR) shall have a maintaining type switch connected such that controller restart can be prevented.

j. Controllers shall have sufficient connections and capabilities to be configurable to any Machinery Control System (MCS) topology in T9070-BR-DPC-010/202-1.

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3.5.4 Cybersecurity measures. Controllers shall have appropriate cybersecurity capabilities to counter threats (accidental or malicious) implementable in accordance with 3.5.4.1. This section does not apply to non-programmable electromechanical controllers.

3.5.4.1 Programmable controllers. Unless otherwise specified (see 6.2), all programmable controllers and their associated circuitry and enclosures shall have the following cybersecurity measures implemented. Additional measures of CNSSI 1253 shall be as specified by the procuring activity:

a. All unused or unnecessary ports shall have the capability of being locked physically or disabled logically. A lockable external port shall be provided for maintenance functions.

b. Unless otherwise specified (see 6.2), programmable parameters shall have an operating range that is modifiable. Exceeding the operating range shall generate an alert to the supervisory network, or locally for standalone controllers.

c. Controllers shall have on-board logging and shall be capable of logging, at a minimum, the most recent 2,000 operating hours. Logs shall have the capability of being secured such that only users with administrative privileges are able to access and edit them. Logs shall, at minimum, include any program changes, successful and unsuccessful authentications, and successful and unsuccessful modifications. Additional logging capabilities, if required, shall be as specified (see 6.2).

d. There shall be no reprogramming capabilities over the connected supervisory network, remotely, or wirelessly.

e. Controllers shall have the capability of being reprogrammed locally with removable media, such as a circuit card or secure digital (SD) card. This media shall be physically located inside the controller enclosure (see 3.5.14).

f. Enclosures shall be capable of being locked physically and shall be provided with fasteners allowing use of lead wire or other tamper evident seals and hasps for use with locks in accordance with A-A-59486 or A-A-59487.

g. Controllers shall be capable of monitoring sensors for failures, which shall generate alerts locally, if so equipped, to the supervisory network.

h. Controllers shall have default parameters and programs stored in write-protected memory.

i. Controllers shall have the capability to create, enable, modify, disable, and remove user accounts. Controllers shall be able to support user accounts based on roles and responsibilities.

(1) Each account shall have rights in accordance with the principle of least privilege.

(2) Factory default authentication credentials shall be removed or changed prior to delivery or installation.

(3) Controllers shall have the ability to support passwords of a minimum of 8 characters up to 16 characters including upper case, lower case, numeric, and special characters.

j. When specified (see 6.2), procedures to restore the controller and its associated sensors shall be provided.

k. When specified (see 6.2), baseline configurations shall be provided. Baseline configurations of controllers shall include, but not be limited to, the following: software packages, version numbers, patch information on operating systems and applications, configuration setting and parameters, network topology, and the logical placement of the components.

l. Controllers shall be configured in accordance with the most current version of the Defense Information Systems Agency (DISA) issued Security Technical Implementation Guide (STIG) and Security Requirements Guide (SRG), which can be found at https://iase.disa.mil/stigs/Pages/index.aspx.

m. When Ethernet ports are used, controllers shall be provided with authentication protocols of the authorized device for the Ethernet port.

n. Controllers shall be capable of validating the syntax and semantics input by users provided from the human-machine interface (HMI) or through the port.

o. Wireless networking capabilities shall be prohibited.

p. Only the software programs necessary to perform controller functions shall be installed, using a deny-all, permit-by-exception (whitelisting) approach.

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3.5.5 Overload protection. Overload protection shall be provided for equipment during starting and running conditions. When an overload protection system requires external power, the power source shall be the same as the served equipment. Resetting, if not automatic, shall be accomplished by means external to the enclosure.

3.5.5.1 Electronic overload protection. Electronic overload relays can be used to protect auxiliary equipment. Electronic overload relays shall provide means to incorporate a physical switch to allow disabling of the LVR mode of operation. Programmable electronic overload relays shall implement the cybersecurity measures of 3.5.4.1. Unless otherwise specified (see 6.2), electronic overload relays shall include, but not be limited to, the following protection features:

a. Cybersecurity capabilities
b. Low voltage protection (LVP)
c. LVR
d. Over current
e. Under current
f. Low voltage
g. High voltage
h. Single phasing
i. Voltage unbalance
j. Current unbalance
k. Phase reversal

3.5.6 Mounting. Parts mounted on panels shall be replaceable without removing the panel.

3.5.7 Emergency run. When specified (see 6.2), magnetic controllers shall be provided with a means of defeating the overload relay for emergency running. This feature may be combined with the overload relay reset or obtained with a separate pushbutton. Manual operation of the emergency run switch shall be required for the duration of the emergency. Emergency run shall not be combined with the start pushbutton. Actuation of the emergency run device and the start device shall be required to initiate emergency run.

3.5.8 Local transfer. When specified (see 6.2), controllers that are intended for use with remote switches shall have a local and remote selector switch. This local and remote selector switch will disconnect the remote control wiring and allow local control. This switch shall prevent a damaged remote switch or defective wiring from disabling the controller. The change in state to local control shall be indicated at the remote control location.

3.5.9 Network communications. Remote configuration or programming capability shall not be permitted by design. All configuration or programming changes and updates shall be performed via a local port that is neither connected nor accessible during normal operation.

3.5.10 HMI. When specified (see 6.2), controllers may utilize touchscreens or other forms or types of HMI (see 3.6.10).

3.5.11 Variable speed drives (VSD). When specified (see 6.2), a VSD can be utilized to provide reduced voltage motor starting capability or variable speed motor control. VSDs shall be in accordance with MIL-PRF-32168.

3.5.12 Controller characteristics. Table I notes the characteristics of controllers that shall be specified (see 6.2). Single-phase operation shall be accomplished utilizing a three-phase controller.

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TABLE I. Controller characteristics.

Controller characteristic	Specifiable choices
Voltage rating	115 VAC
	440 VAC
	24 VDC
	115 VDC (nominal) shall operate between 80 and 110 percent of rated voltage
	230 VDC (nominal) shall operate between 80 and 110 percent of rated voltage
	250 VDC (nominal) range 180 to 355 VDC
Current rating	Current rating shall be as required by the applicable auxiliary equipment use.
Magnetic controller type: Across the line	Single-speed
	Multi-speed
Magnetic controller type: Reduced voltage	Autotransformer
	Primary resistor
	Primary reactor
	Resistor (DC)
	Wye delta
Protection	LVP
	LVR
	Overload relay
	Temperature sensor actuated
Function	Equipment starting
	Motor starting, reversing, and speed selection
	Equipment stopping
	Miscellaneous
Manual switches	Pushbutton
	Rotary selector
	Drum
	Float

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TABLE I. Controller characteristics – Continued.

Automatic switches	Pressure
	Thermostatic
	Limit
	Proximity
	Photoelectric
Switch functions	Actuate a magnetic controller
	Shift local to remote
	Shift remote to local

3.5.13 Continuous amperage ratings. Controllers shall be provided in the sizes, and shall conform to the current rating, shown in table II.

3.5.14 Enclosures. Enclosures shall be in accordance with MIL-DTL-2036 and shall utilize locks in accordance with A-A-59486 or A-A-59487. Sizes shall be in accordance with table II. If a particular controller requires conduit, sensor cabling, or wiring that exceeds the size for the rated amperage, the next larger size shall be used. When possible, the NEMA and Underwriters Laboratories (UL) enclosure sizes listed may be used.

TABLE II. Basic continuous amperage ratings and enclosure dimensions of controllers.

Size (MIL-DTL-2212)	PIN code	AC Amperage Rating	DC Amperage Rating
N/A	A	Per UL	Per UL
N/A	B	Per NEMA	Per NEMA
N/A	C	Per NEMA	Per NEMA
0	D	18	15
1	E	27	25
2	F	45	50
3	G	90	100
4	H	135	150
5	I	270	300
6	J	540	600
7	K	810	900

3.5.15 Equipment capacitance and EMI filters. Where EMI filters are required, line-to-line filters shall be used. AC line-to-ground capacitance shall be in accordance with MIL-STD-1399-300 and MIL-STD-461.

3.5.16 Pickup and dropout voltage. Pickup and dropout voltages for controller components shall be in accordance with MIL-DTL-2212.

3.6 Construction. Unless otherwise specified herein (see 6.2), auxiliary controller construction shall be in accordance with MIL-DTL-2212.

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3.6.1 Duty. Unless otherwise specified (see 6.2), controllers shall be constructed for continuous duty and general purpose service. Controllers for special duty, such as jogging or plugging (induction braking), constant-current, multi-speed motors, lighting, or transformer loads, may require special ratings.

3.6.2 Load isolation. Unless otherwise specified (see 6.2), opening of the line contactors shall isolate all portions of the load from the line except control circuits and the incoming power terminals. A discharge circuit shall be provided for shunt field winding DC controllers.

3.6.3 Remote device circuits. Control circuit leads from controllers to remote devices shall be protected against damage from short circuits. Leads wired in series with current limiting parts, such as coils and resistors located within the enclosure, are considered to be adequately protected. Leads not protected by current limiting parts within the enclosure shall be fused. Unless otherwise specified (see 6.2), remote indicating and monitoring circuits shall be independently fused such that a short circuit in these circuits will not prevent the controller from performing its function. Fuse sizes shall be in accordance with MIL-DTL-917. When fuses larger than 10-ampere capacity are required, 15- or 20-ampere fuses shall be used as appropriate.

3.6.4 Wiring. Permanent internal wiring shall be in accordance with MIL-DTL-917.

3.6.5 Fiber optics. In accordance with T9070-BR-DPC-010/202-1, fiber optic cable shall not be used to connect a controller to the served equipment. Fiber optics may be used inside the enclosure and to connect to the supervisory network in accordance with MIL-STD-2042.

3.6.6 Interlocks. Multi-speed and reversing controllers shall use electrical and mechanical interlocks to prevent attempted operation in more than one mode.

3.6.7 Contactors and relays. Contactors and relays shall be in accordance with MIL-DTL-2212.

3.6.8 Switches. Switches shall be in accordance with MIL-DTL-15291, MIL-DTL-18396, and MIL-DTL-21604, or the applicable requirements of this specification. Each position of manual switches, other than spring return switches, shall have definite positioning that can be felt by the operator. Switch identification plates shall clearly indicate each position. Switches shall be mountable in any position.

a. Pressure switches. Pressure and differential pressure switches shall be in accordance with MIL-DTL-2212.

b. Temperature switches. Temperature switches shall be in accordance with MIL-DTL-2212.

c. Mechanical limit switches. Where practical, the switch assemblies utilized shall be in accordance with MIL-PRF-8805 and MIL-DTL-2212.

d. Solid-state proximity switches. Solid-state proximity switches shall be in accordance with MIL-PRF-24711 and MIL-DTL-2212.

e. Solid-state photoelectric switches. Solid-state photoelectric switches shall be in accordance with MIL-DTL-2212.

f. Pushbutton switches. Pushbutton switches shall be in accordance with MIL-PRF-8805 and ASTM F1166. Pushbutton switches shall be rated for the same or higher operating voltage and dielectric withstand voltage as the controller control circuit. The contact current make and break rating shall be matched to the device being controlled. Pushbutton switches shall be capable of the same tightness requirements as the controller enclosure in which they are installed. Pushbutton switch assemblies shall be installed in the controller and shall not fail when the controller is subjected to verification testing requirements.

3.6.9 Transformers. Transformers shall meet requirements of MIL-DTL-2212.

3.6.10 HMI. HMIs shall meet the following requirements:

a. All HMIs shall be in accordance with the human engineering requirements of ASTM F1166 and MIL-STD-1472 when installed and operated in its intended location.

b. HMI touchscreens, if used, shall be of a flat panel design utilizing an active matrix thin film transistor liquid crystal anti-glare color visual display. In order to activate the desired event, the operator shall touch the appropriate touchscreen control icon. When the operator discontinues contact with the touchscreen control, the “dead-man” or return-to-center feature specified for that control shall activate.

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c. HMI visual displays shall incorporate brightness controls, anti-glare features, and display color schemes that enable the display to be read and operated when in direct sunlight as well as in low-light conditions within ships.

d. HMI screens, if used, shall be capable of displaying graphical user interface (GUI) displays such as pushbuttons, selector switches, numeric keypads, diagnostic indicators, message displays, bar graphs, text displays, and list indicators.

e. All HMIs shall have a life of at least 100,000 hours.

f. All HMIs shall be capable of operating from either 24-VDC or 115-VAC power.

g. All HMIs shall contain non-volatile flash memory for the storage of the HMI program.

h. All HMIs shall, unless otherwise specified (see 6.2), interface with the controller via an open architecture network communication protocol.

i. All HMIs shall be provided with a hinged or removable cover to prevent inadvertent display screen contact by personnel or objects. The cover shall not prevent personnel from reading the display screen when the cover is positioned to protect the screen. The cover shall not be used as the primary source of EMI protection in order to comply with MIL-STD-461 (see 3.5.15).

j. All HMIs shall not be used as the emergency stop pushbutton or run/stop selector switch.

k. All HMIs shall emulate the detent, spring return, and “dead-man” characteristics specified for those control functions where HMIs are used in lieu of electromechanical selector or pushbutton switches.

l. All HMIs shall incorporate self-diagnostics. HMI fault status and diagnostic information shall be accessible through the HMI display.

m. All HMIs shall have a reset button that reboots the HMI when depressed

n. All HMIs shall be equipped to meet cybersecurity requirements of 3.5.4.

3.6.11 Space heaters. When specified (see 6.2), controllers in watertight or submersible enclosures shall be provided with space heaters to prevent condensation.

3.6.12 Indicator lights. If indicator lights are required (see 6.2), AC indicator lights shall be in accordance with MIL-DTL-3661 and MIL-L-3661/65 for 450- and 120-volt service. DC indicator lights shall be in accordance with MIL-L-3661/63 or shall be a type acceptable to NAVSEA. LED indicator lights that meet the requirements of A-A-59781 may be utilized provided the LED indicator lights have been subjected to all qualification requirements for the controller.

3.6.13 Fuses and fuseholders. Fuses shall be in accordance with MIL-DTL-2212.

3.6.14 Resistors and rheostats. Resistors and rheostats shall be in accordance with MIL-DTL-15109.

3.6.15 Terminals and terminal boards. Power current carrying studs, screws, and other current carrying components of power connections shall be constructed of nonferrous materials. Control current carrying components or connections shall be constructed of either nonferrous materials or silver-plated ferrous materials. Where practical, terminal boards shall be in accordance with A-A-59125. Construction materials shall be durable and corrosion-resistant.

3.6.16 Resilient mounting. When the momentary opening of closed contacts or the momentary closing of open contacts of components within a controller as specified in 3.4.2 is not acceptable, resilient mounts may be utilized in accordance with S9073-A2-HBK-010. Controllers utilizing resilient mounts shall be able to withstand the shock and vibration tests specified in 4.6.11 and 4.6.12 with the resilient mounts installed as specified (see 6.2).

3.6.17 Insulation resistance. Insulation resistance shall be not less than 10 megohms when tested as specified in 4.6.4.

3.6.18 Dielectric. Controllers, contactors, and associated switching devices shall be able to withstand the dielectric tests specified in 4.6.5 without electrical breakdown such as corona, flashover, spark over (surface discharge), or breakdown (puncture).

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3.6.19 Terminal strength. The terminals of the contactors and control relays shall withstand the terminal strength requirements of MIL-STD-202 when tested as specified in 4.6.7.

3.7 Endurance. Endurance requirements for contactors, switches, relays (except relays in accordance with MIL-R-19523), and overload relays shall be in accordance with MIL-DTL-2212.

3.8 Reliability. Controllers shall be designed such that the mean time between failure is a minimum of 100,000 hours of operation, where failure is defined as any condition that prevents the controller from meeting any requirements herein.

3.9 Maintainability. The mean time to repair shall be not greater than 3 hours, and the maximum time to repair shall be not greater than 6 hours. The mean maintenance time per month shall be not greater than 30 minutes.

3.10 Safety. Equipment shall be constructed to ensure safety to operating and maintenance personnel in accordance with MIL-DTL-917. MIL-HDBK-454 provides additional guidance on safety features of controllers. Personnel protection from hazardous voltages should follow the guidelines established in S9086-KC-STM-010/300.

3.11 Finish. Exterior finish shall be in accordance with MIL-DTL-2036. Parts not constructed of corrosion-resistant material, or having a corrosion-resistant treatment in accordance with MIL-DTL-917, shall be painted as specified therein. Touch-up painting is permitted for marks or scratches due to assembly, testing, or other handling.

3.12 Identification plates and diagrams. Each controller shall have an identification plate containing the manufacturer's name, catalog number, national stock number (NSN) (when specified [see 6.2]), and PIN (see 1.3). The inside of the enclosure shall include a connection diagram, a schematic diagram, a description of operation, and a table listing overload heaters, fuses, circuit breakers, coils, and reactors (if used). The information shall be printed by a process that is nonfading and protected by transparent plastic. This information shall be legible and shall be as large as is practical based on the size of the controller enclosure, but shall not exceed 28 by 43.2 cm (11 by 17 inches), foldable to 21.6 by 28 cm (8.5 by 11 inches). All internal wire and cable shall be clearly identified and marked in accordance with MIL-DTL-917.

3.13 Workmanship. Controllers, including all parts and accessories, shall be free from cracked or displaced parts, burrs, sharp edges, other defects that will detrimentally affect life and serviceability, and shall meet the requirements for neatness and thoroughness of soldering in accordance with J-STD-001, class 3. Particular attention shall be given to marking of parts and subassemblies, wiring, welding and brazing, plating, riveting, finishes, machine operations, and screw assemblies.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in MIL-DTL-917.

4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory acceptable to NAVSEA. Qualification inspection shall consist of the examination and tests specified in table III and shall be conducted in the order specified in 4.3.3.

4.3.1 Change approval. A change in material, parts, production processes, production equipment, baseline programming, or manufacturing facilities used in the manufacture of the equipment, which have been qualified, shall require written approval of NAVSEA. Incorporation of any changes which have not been so approved shall require requalification of item in question.

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TABLE III. Qualification and conformance inspections.

Inspection	Requirement(s)	Qualification	Conformance
Examination, and general operation	3.3, 3.5, 3.5.1, 3.5.2, 3.5.3, 3.5.6, 3.5.7, 3.5.8, 3.5.11	4.5	4.5
Environmental	3.4	4.6.1	4.6.1
Noise	3.4.1	4.6.2	---
Enclosures	3.5.14	4.6.3	---
Insulation resistance	3.6.17	4.6.4	---
Dielectric	3.6.18	4.6.5	4.6.5
Pickup and dropout voltage	3.5.16	4.6.6	---
Terminal strength	3.6.19	4.6.7	---
Overload protection	3.5.5	4.6.8	---
Electric service	3.4.6	4.6.9	---
Voltage ramp 1/	3.4.6	4.6.9.1	---
Endurance	3.7	4.6.10	---
Shock	3.4.2	4.6.11	---
Vibration	3.4.3	4.6.12	---
EMI	3.4.4	4.6.13	---
Inclination	3.4.5	4.6.14	---
Cybersecurity	3.5.4, 3.5.9	4.5	4.5
Touchscreen	3.5.10 and 3.6.10	4.5	---
Controller characteristics	3.5.12	4.5	---
Continuous amperage ratings	3.5.13	4.5	---
EMI filters	3.5.15	4.5	---
Construction	3.6, 3.6.1 through 3.6.9, and 3.6.11 through 3.6.16	4.5	---
Reliability	3.8	4.6.15	---
Maintainability	3.9	4.6.15	---
Safety	3.10	4.5	4.5
Finish	3.11	4.5	4.5
Identification plates and diagrams	3.12	4.5	4.5
Workmanship	3.13	4.5	4.5

NOTE:
1/ Test applies to qualification of devices to be operated under the condition of voltage ramp when specified.

4.3.2 Samples for qualification. One to five samples of each controller for which qualification is desired shall be submitted for qualification testing.

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4.3.3 Order of tests. Unless otherwise specified (see 6.2), the tests for general operation, effectiveness of enclosure, insulation resistance, dielectric, pickup and dropout, terminal strength, voltage variation, overload protection, and voltage ramp (see 4.6.3 through 4.6.9.1) shall precede and follow shock and vibration tests (see 4.6.11 and 4.6.12). The endurance test (see 4.6.10) shall precede the shock test (see 4.6.11).

4.3.4 Disposition of qualification units. Samples subjected to qualification inspection shall be considered consumed and nondeliverable as specified (see 6.2).

4.3.5 Failures. Failure of any sample to meet the requirements specified herein shall be cause for disapproval of product qualification.

4.4 Conformance inspection. Conformance inspection shall be performed on each controller furnished (100 percent inspection) and be as specified in table III.

4.5 Examination, general operation, and cybersecurity. Equipment shall be examined to determine that the material, safety to operating personnel, construction, and workmanship are in accordance with the requirements of this specification. These examinations shall not require disassembling the unit in such a manner that its performance, durability, or appearance would be affected. The general operation shall include energizing and operating the equipment to ascertain that the required operating features of the unit function properly. In particular, it shall be determined that the required sequence of operation has been attained and that the protective devices, including cybersecurity measures, operate as required. External devices shall be simulated as necessary to perform operational tests. Specific electrical loads are not required, but at least one set of each type of contact on every device shall be monitored to check the opening and closing performance. For both qualification and conformance inspections of programmable controllers and associated devices having networking/interfacing capabilities (such as electronic overloads and HMIs, if installed), the following additional tests are required:

a. Test and demonstrate all executable commands and status requests for all installed network interface protocols (see 3.5.4 and 3.5.9).

b. Test and demonstrate the operation of all inputs and outputs via both local and network protocol commands as appropriate for all possible configurations of which the device is capable (e.g., single-speed, multi-speed, reversing).

c. Test and demonstrate all configuration types of which the device is capable (e.g., single-speed, multi-speed, reversing).

d. Test and demonstrate proper operation of the LVP and LVR modes for all available configurations (see 3.5.12).

e. Demonstrate the ability to display the firmware version locally or via network interface if applicable.

f. Demonstrate the capability and operability of cybersecurity measures of 3.5.4.

g. For touchscreen HMIs, if installed, demonstrate compliance with the requirements of 3.6.10.

4.5.1 Voltage variation. Controllers shall be connected to a voltage supply that can be varied from 0 to 110 percent of the controller rated input voltage to determine conformance to 3.4.1 and 3.4.2. The controller shall not malfunction due to the input voltage varying between 80 and 110 percent of rated value. An examination and general operation test shall be performed at the low and high points to determine that the controller operates properly.

4.6 Tests. All tests shall be conducted at an ambient temperature range between 10 and 40 °C (50 and 104 °F). Equipment shall be operated at nominal voltage and frequency during the tests and mounted in a fashion to simulate shipboard conditions, including sway bracing when required. Tests, when practicable, shall be performed continuously without interruption.

4.6.1 Temperature and humidity. Temperature and humidity tests shall be performed for qualification. For conformance, this inspection shall be performed on the first production run only. Temperature and humidity tests are not required on subsequent production runs unless a change is made in design. Temperature test shall be conducted in accordance with MIL-STD-810, high temperature and low temperature tests. For high temperature, the specific test shall be “Basic Hot” for three cycles. For low temperature, the specific test shall will be “Basic Cold” with the temperature range for the test relaxed to -2 to 0 °C. Humidity test shall be conducted in accordance with

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MIL-STD-810, humidity tests (cycle B1, seven cycles).

4.6.2 Noise. Noise tests shall be conducted in accordance with MIL-STD-1474 to determine conformance to the requirements specified in 3.4.1 in the normal operating mode.

4.6.3 Effectiveness of enclosure. Enclosures shall be subjected to the applicable test conditions specified in MIL-DTL-2036 and NEMA 250 to determine conformance to 3.5.14.

4.6.4 Insulation resistance. The measuring instrument used shall have an open circuit voltage of 500 VDC ±10 percent. Measurements shall be made between all mutually insulated points and between insulated points and ground (see 3.6.17).

4.6.5 Dielectric. Dielectric tests shall be conducted in accordance with NEMA ICS 1 with a test voltage (rms) of 1,000 volts plus (2 times nominal voltage rating) for controllers and all other devices in the associated enclosure.

4.6.6 Pickup and dropout voltage. Controller contacts, relays and solid-state devices shall be connected to a voltage supply that can be varied from 0 to 110 percent of the rated voltage to determine pickup and dropout voltages. Performance shall be in accordance with 3.5.16. The pickup and dropout voltage test for qualification shall be conducted with the contactor or relay operating at maximum temperature, unless the tests are being performed as part of another test.

4.6.7 Terminal strength. The terminals of the contactors and control relays shall be tested in accordance with torque test method 211A of MIL-STD-202 (see 3.6.19).

4.6.8 Overload protection. Overload relays shall be tested at an ambient temperature of 50 °C (122 °F) and at 110, 125, 150, 300, and 500 percent of rated current. Resetting, if not automatic, shall be accomplished by means external to the enclosure.

4.6.9 Electrical service conditions. Controllers shall be tested in accordance with MIL-DTL-2212 for steady-state voltage and frequency, transient voltage and frequency, spike voltage, power interruption, leakage current, output leakage current, and output voltage drop.

4.6.9.1 Voltage ramp rate test. When specified (see 6.2), equipment shall be tested at the minimum ramp rate and at a voltage step (0 – maximum voltage instantaneously) as follows:

DC: 54 volts per second and voltage step

AC: 900 volts per second and voltage step

4.6.10 Endurance. Endurance tests shall be conducted between 10 and 20 operating cycles per minute, except tests making and breaking 600 to 400 percent of rated current shall be conducted at 6 operating cycles per minute (closed 1 second, open 9 seconds). Tests making and breaking 600 to 400 percent of rated current shall be performed continuously without interruption. Tests breaking 100 percent rated current may be stopped and restarted for convenience provided each operational test period lasts a minimum of 6 hours. A malfunction causing shutdown or out of specified limit performance during this period shall constitute a failure.

4.6.11 Shock. Shock tests shall be conducted in accordance with grade A, class I, type A shock test of MIL-DTL-901 to determine conformance to 3.4.2. A separate unit may be used for each set of shock blows. MIL-DTL-901 bulkhead fixtures shall be used (fixture 4A mounting for lightweight tests). The following are definitions of failure:

a. Breaking or cracking of any part, including mounting bolts, or material around the mounting holes that will affect device function.

b. Failure to pass insulation resistance tests.

c. Failure to pass pickup and dropout tests for individual contactors and relays.

d. Failure to pass voltage variation tests for controllers.

e. Failure to pass dielectric tests.

f. Failure to pass the visual examination.

g. Appreciable distortion of any parts, including the enclosure and framework.

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h. Failure to meet the shock requirements specified in 3.4.2.

i. Any change of state of a controller, contactor, or associated switching device.

j. Causes electrical safety hazard to personnel due to shorts, arcing, or loose or damaged wiring.

4.6.11.1 Monitoring during shock test. On shock tests of components, each contact, including AC and DC line contacts as applicable, shall be monitored. Electronically-based control circuit devices shall be monitored with all inputs and outputs operating to ensure that they do not change state. Each network protocol port that is available for the device shall be monitored during the shock tests. Network status and operational commands shall be continuously executed to the device to ensure that network communications are properly operating during the entire test.

4.6.11.2 Post-shock tests and examinations. Upon completion of shock testing, the following tests and examinations shall be performed:

a. Visual examination. Equipment shall be carefully examined to ascertain any mechanical damage. Equipment shall be disassembled as necessary to detect any damage.

b. Examination, general operation, and cybersecurity. An examination, general operation, and cybersecurity inspection shall be performed as specified in 4.5.

c. Insulation resistance. Insulation resistance shall be measured as specified in 4.6.11 and shall meet the requirements of 3.6.17.

d. Post-shock dielectric. A dielectric test shall be conducted as specified in 4.6.12 except that the test voltage applied will be 65 percent of the specified value and will be applied for 5 seconds. Performance shall meet the requirements of 3.6.18.

e. Pickup and dropout voltage. When tested individually, contactors and relays, including time delay relays, shall be tested as specified in 4.6.10, except at ambient temperature between 10 and 40 °C (50 and 104 °F), and shall meet the requirements of 3.5.16.

f. Voltage variation. Controllers shall be tested as specified in 4.5.1 and shall meet the requirements of 3.4.1 and 3.4.2.

4.6.12 Vibration. Before vibration testing, the tests of 4.5 shall be performed. The equipment shall then be subjected to the type I vibration tests specified in MIL-STD-167-1 and shall meet the requirements of 3.4.3. The vibration test frequency shall be 4 to 33 Hz. Equipment which will be exposed to frequencies above 33 Hz shall be tested up through 50 Hz. Equipment shall not fail to function in any operational mode, including energized and de-energized conditions, during the vibration. Specific electrical loads are not required, but at least one set of each type of contact on every device shall be monitored to check the opening and closing performance. Each set of contacts shall be monitored. Programmable controller vibration tests shall be conducted with all inputs and outputs operating or powered and monitored to ensure they do not change state. After the vibration test, the tests of 4.5 shall be repeated.

4.6.13 EMI. The emission and susceptibility tests specified in MIL-STD-461 shall be performed to determine electromagnetic compliance with 3.4.4. In addition, the energized device shall be subjected to a high current field of 500 amperes per foot. The field shall be applied in each axis (top, bottom, left, right, front, and back). During this testing, the device shall be constantly polled via the network interface and all input and outputs monitored. There shall be no error in the data stream or unexpected change in state of the device.

4.6.14 Inclination. Equipment shall pass the tests of 4.5 when inclined from the bulkhead mounted orientation at the maximum angles specified in 3.4.5. Tests shall be conducted at nominal voltage and frequency. Controllers shall be subjected to the voltage variation test specified in 4.5.1 at each of the four test positions.

4.6.15 Reliability and maintainability. The controller shall meet the requirements for reliability specified in 3.8. The validation procedures in MIL-HDBK-217 may be used. The controller shall meet the requirements for maintenance and repair specified in 3.9. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during qualification testing. The demonstration procedures in MIL-HDBK-470 may be used.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Controllers covered by this specification are intended for use in naval surface ships and submarines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Environmental conditions other than specified herein (see 3.4).
c. Shock classification for associated devices other than specified (see 3.4.2.c and d).
d. Electrical service conditions, when specified (see 3.4.6).
e. Controller design type (programmable or non-programmable) (see 3.5).
f. Control circuit voltage, if not same as the power circuit (see 3.5.3.b).
g. Programmable controllers if not specified (3.5.4.1).
h. Programmable parameter limits other than ±25 percent, if applicable (see 3.5.4.1.b).
i. Logging requirements, if different from those specified (see 3.5.4.1.c).
j. Controller restoration procedures (see 3.5.4.1.j).
k. Controller baseline configuration (see 3.5.4.1.k).
l. Overload protection, if not as specified (see 3.5.5.1).
m. Emergency run feature, if required (see 3.5.7).
n. Local transfer switch for vital function, if required (see 3.5.8).
o. Human machine interface, if required (see 3.5.10).
p. Variable speed drive, if required (see 3.5.11).
q. Controller characteristics (see 3.5.12).
r. Construction requirements, if other than as specified (see 3.6).
s. Duty, when other than continuous (see 3.6.1).
t. Load isolation requirement, if other than as specified (see 3.6.2).
u. Remote device circuits, if other than as specified (see 3.6.3).
v. Touchscreen HMI controller interface requirements, if other than as specified (see 3.6.10.i).
w. When space heaters are required in enclosures (see 3.6.11).
x. When indicator lights are required (see 3.6.12).
y. Requirements for resilient mounting of equipment (see 3.6.16).
z. When NSN is required on the identification plate (3.12).
aa. Inspection conditions, if other than those specified in MIL-DTL-917 (see 4.2).
ab. Order of tests, if other than as specified (see 4.3.3).
ac. Disposition of sample units (see 4.3.4).

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ad. When voltage ramp rate test is required (see 4.6.9.1).

ae. Packaging requirements (see 5.1).

af. Recommended documentation and programs (see 6.3).

6.3 Recommended documentation and programs. Engineering drawings, certifications, software and hardware configuration descriptions, and software programs to perform controller functions (installed as applicable) should be requested if necessary (see 6.2).

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32613 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Definitions.

6.5.1 Ambient temperature. Ambient temperature is the temperature of the medium such as air, water, or earth into which the heat of the equipment is dissipated. For self-ventilated equipment, the ambient temperature is the average temperature of the air in the immediate neighborhood of the equipment. For air or gas-cooled equipment with forced ventilation or secondary water-cooling, the ambient temperature is taken as that of the in going air or cooling gas. For self-ventilated enclosed (including oil-immersed) equipment considered as a complete unit, the ambient temperature is the average temperature of the air outside of the enclosure in the immediate neighborhood of the equipment.

6.5.2 Associated switching device. Associated switching devices are devices that are used in conjunction with the main line contactor(s) of a controller to control the operation of the connected load and may be mounted separately from the controller. These include control and overload relays, limit switches, pushbutton stations, pressure switches, selector switches, level switches, and others.

6.5.3 Automatic. Automatic means self-acting, operating by its own mechanism when actuated by some impersonal influence such as, for example, a change in current strength, pressure, temperature, or mechanical configuration.

6.5.4 Auxiliary systems. Auxiliary systems include a broad spectrum of machinery and equipment aboard ship, comprising those systems which are not the main propulsion or principal electric plant of the ship. This includes, but is not limited to, secondary propulsion, emergency power, distilling and reverse-osmosis water plants, pumps, heaters, HVAC and refrigeration equipments, compressors, galley, laundry and shop equipments, and deck machinery.

6.5.5 Contact chatter. Contact chatter is a transient condition characterized by the repetitive bouncing (make and break) of mating electrical contacts. Each contact bounce is usually denoted by transient voltage excursions equal to or greater than 90 percent of the applied circuit voltage.

6.5.6 Contactor. A contactor is a device for repeatedly establishing and interrupting an electric power circuit.

6.5.7 Contactor, magnetic. A magnetic contactor is a contactor actuated by electromagnetic means.

6.5.8 Contacts. Contacts are conducting parts which coact to complete or to interrupt a circuit.

6.5.9 Contacts, auxiliary. Auxiliary contacts of a switching device are contacts in addition to the main circuit contacts and function with the movement of the latter.

6.5.10 Contacts, line. Line contacts are contacts on a contactor or relay used solely for controlling the power to a motor or controlled device.

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6.5.11 Contacts, normally open (NO) and normally closed (NC). Normally open and normally closed signify the position taken by contacts when the device is de-energized. The de-energized condition of a manual device is the off position.

6.5.12 Control circuit. The control circuit of a control apparatus or system is the circuit which carries the electrical signals directing the performance of the controller but does not carry the main circuit power.

6.5.13 Control, local. Local control or device is a control function or device which provides for initiation or change of a control function from within or directly upon the controller enclosure.

6.5.14 Control relay. A control relay is a device for repeatedly establishing and interrupting an electric circuit. A control relay is usually employed to actuate other relays or contactors in the same or associated circuits.

6.5.15 Control, remote. Remote control or device is a control function or device which provides for initiation or change of a control function from a remote point.

6.5.16 Controller, full-magnetic. A full-magnetic controller is an electric controller having all of its basic functions performed by devices which are operated by electromagnets. Basic functions usually refer to acceleration, retardation, line closing, reversing, and so forth.

6.5.17 Corrosion-resistant. Corrosion-resistant means constructed, protected, or treated so that corrosion will not exceed specified limits under specified test conditions.

6.5.18 Disconnection. Disconnection is the opening of a sufficient number of conductors to prevent current flow.

6.5.19 Dropout voltage or current. The dropout voltage or current of a magnetically operated device is the voltage or current at which the device will return to its de-energized position.

6.5.20 Duty, continuous. Continuous duty is a requirement of service which demands operation at a substantially constant load for an indefinitely long time.

6.5.21 Duty, intermittent. Intermittent duty is a requirement of service which demands operation for alternate intervals of (1) load and no-load, (2) load and rest, or (3) load, no-load, and rest, with such alternate intervals being specified.

6.5.22 Duty, varying. Varying duty is a requirement of service which demands operation at wide variations of loads and intervals of time.

6.5.23 Emergency run. Emergency run is a condition in which the protective devices that normally interrupt a control circuit are temporarily and manually overridden by some means and cause the circuit function to be maintained.

6.5.24 Enclosure. An enclosure is a surrounding case designed to protect:

a. Personnel against accidental contact with the enclosed electrical devices, and

b. Internal devices against specified external conditions.

6.5.25 Fuse. A fuse is an over-current protective device with a circuit opening fusible member which is heated and severed by the passage of over-current through it.

6.5.26 Grounded parts. Grounded parts are parts which are connected in an intentionally grounded manner.

6.5.27 Input and output characteristics. Input devices accept signals from the process elements and translate them into signals which can be used within the controller. Output devices convert controller signals into external signals used to control the process.

6.5.28 Interlock. An interlock is a device actuated by the operation of some other device with which it is directly associated to govern succeeding operations of the same or allied devices. Interlocks may be either electrical or mechanical.

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6.5.29 Isolation. Isolation is the opening of all conductors connecting to the power source.

6.5.30 Jog. Jog is a control function which permits the momentary operation of a drive for the purpose of accomplishing a small movement of the driven machine.

6.5.31 Limit, current. Current limit is a control function which prevents a current from exceeding its prescribed limits. Current limit values should be expressed as a percent of rated load value.

6.5.32 Low voltage protection (LVP). LVP is the effect of a device which, on the reduction or failure of voltage, causes and maintains the interruption of power to the main circuit until the circuit is restarted manually.

6.5.33 Low voltage release (LVR). LVR is the effect of a device which, on the reduction or failure of voltage, causes the interruption of power to the main circuit, but allows the automatic reestablishment of the main circuit on return of voltage.

6.5.34 Naval control system. System containing analog or digital electronic devices, or both, (possibly computers) containing output signals that, either directly or indirectly, effect change external to the system and control a physical process. This includes systems designated as MCS; supervisory control and data acquisition systems; warfare, navigation, and hull, mechanical, and electrical systems; and industrial control systems. Information technology systems are excluded. (Definition per: SCG ID# 10-040 U.S. Navy Afloat Control Systems Cyber Security Classification Guide [SCG])

6.5.35 Open momentarily, or momentary opening. Where applied in this specification, these expressions mean that an interval of contact physical opening may occur during high-impact shock.

6.5.36 Pickup and seal voltage. The pickup and seal voltage of a magnetically operated device is the minimum voltage at which the device moves from its de-energized into its fully energized position.

6.5.37 Pickup voltage or current. The pickup voltage or current of a magnetically operated device is the voltage or current at which the device starts to operate.

6.5.38 Plugging. Plugging is a method of braking used in induction motors. Plugging involves interchanging the supply to two of the stator phase windings. This method is used in applications which require immediate stop applications. When the phase supply is reversed, torque is produced in the opposite direction. This leads to braking of the electric motor.

6.5.39 Principle of least privilege. The concept of limiting access to only the information and resources that are necessary for its legitimate purposes.

6.5.40 Programmable logic controller (PLC). A PLC is a digitally operating electronic apparatus which uses a programmable memory for the internal storage of instructions for implementing specific functions such as logic, sequencing, timing, counting, and arithmetic to control, through digital or analog input and output modules, various types of processes.

6.5.41 Protection, overload. Overload protection is the effect of a device operative on excessive current, but not necessarily on short circuit, to cause and maintain the interruption of current flow to the device governed.

6.5.42 Pushbutton. A pushbutton switch (pushbutton) is a master switch having a manually operable plunger or button for actuating the switch.

6.5.43 Rating, continuous. Continuous rating is the rating which defines the substantially constant load which can be carried for an indefinitely long time.

6.5.44 Rating of a controller. The rating of a controller is an arbitrary designation of an operating limit. It is based on the power governed and on the duty and service required. A rating is arbitrary in the sense that it must necessarily be established by definite fixed standards and cannot, therefore, indicate the safe operating limit under all conditions which may occur.

6.5.45 Relay, magnetic control. A magnetic control relay is a relay which is actuated by electromagnetic means. When not otherwise qualified, the term refers to a relay intended to be operated by the opening and closing

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of its coil circuit, and having contacts designed for energizing and deenergizing the coils of magnetic contactors or other magnetically operated devices.

6.5.46 Relay, magnetic overload. A magnetic overload relay is an overcurrent relay in which the electrical contacts are actuated by the electromagnetic force produced by the full or partial load current.

6.5.47 Relay, overload. An overload relay is an overcurrent relay which functions at a predetermined value of overcurrent to cause disconnection (see 6.5.19) of the load from the power supply. An overload relay is intended to protect the load (for example, motor armature) of its controller and does not necessarily protect itself.

6.5.48 Reset. To reset means to restore a mechanism, storage, or device to a prescribed state.

6.5.49 Resistant (used as a suffix). Resistant, used as a suffix, means constructed, protected, or treated so that the apparatus will not be damaged when subjected to the specified material or conditions for a specified time.

6.5.50 Resistor. A resistor is a device used to introduce resistance into an electric circuit.

6.5.51 Reversing. Reversing is a control function which permits changing operation of the drive from one direction to the other.

6.5.52 Rheostat. A rheostat is an adjustable resistor constructed so that its resistance may be changed without opening the circuit in which it is connected.

6.5.53 Sensing element. Sensing element is a primary measuring device for detecting either the absolute or variable pressure or temperature.

6.5.54 Service of a controller. The service of a controller is the specific application in which the controller is to be used, for example:

a. General purpose.

b. Definite purpose:

(1) Crane and hoist.

(2) Elevators.

(3) Machine tool.

6.5.55 Supervisory network. See 6.5.34 above.

6.5.56 Switch. A switch is a device for making, breaking, or changing the connections in an electric circuit.

6.5.57 Temperature rise. Temperature rise refers to the increase in temperature above ambient, measured in degrees Celsius, which will be reached by a device during its normal operation.

6.5.58 Tests, dielectric. Dielectric tests are tests which consist of the application of a voltage higher than the rated voltage for a specified time for the purpose of determining the adequacy against breakdown of insulating materials and spacings under normal conditions.

6.5.59 Time delay. Time delay means that a time interval is purposely introduced in the performance of a function.

6.5.60 Ultimate tripping current. Ultimate tripping current is the minimum value of continuously applied current that will cause an overload relay to operate (trip).

6.5.61 Ventilated. Ventilated means constructed to provide for the circulation of external air through the enclosure to remove heat, fumes, or vapors.

6.5.62 Watertight. Watertight means the equipment is constructed to exclude the entry of water while submerged in water to a certain depth.

6.5.63 Waveform, output. Output waveform is the graphical representation of the output during one pulse interval.

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6.5.64 Weathertight. Weathertight means constructed to exclude water applied in the form of a hose stream under specified test conditions.

6.6 Subject term (key word) listing.

Cybersecurity

PLC

Programmable

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 6110-2018-001)

Review activity:
DLA – GS

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